

Weather-Label-Free Salient Object Detection: A Strong Baseline and a Null Result for Spatial Mixture-of-Experts

Dr. Sahadeb Shit Bapatu Avinash Reddy Lalith Madhav Chundi Anudeep Muppalla

Department of Computer Science and Engineering

Abstract— Salient object detection degrades under adverse weather. We ask whether spatial token-level mixture-of-experts (MoE) routing, used without weather labels, helps. The answer is that it does not: on WXSOD the model reaches MAE 0.0192 on 1,500 synthetic and 0.0168 on 554 real test images, competitive with weather-supervised methods. A full ablation, however, shows the mixture contributes nothing measurable: removing it, routing at one scale instead of three, or replacing routed experts with a dense expert all leave real-test MAE within 0.0004 of the full model, inside the seed-variance floor; entropy sits at 99.8–100% of its ceiling in every arm, and random routing costs under 0.0005 MAE. The numbers come from the backbone and the training budget – 14 epochs instead of 8 improves MAE more than any architectural change we tested – not from routing, and we report that dissociation rather than over-crediting the MoE.

Index Terms— salient object detection, mixture-of-experts, adverse weather, sparse routing, cross-attention decoder

I. INTRODUCTION

SOD identifies visually prominent regions as pixel-wise masks, but performance degrades under adverse weather. Two strategies address this: (1) weather-label-conditioned models—NIFM [1]—require metadata at inference; (2) MoE for weather restoration—WM-MoE [2]—targets pixel regression, not segmentation. To our knowledge, existing MoE-SOD methods primarily explore modality-, scale-, or task-level routing for multi-modal inputs rather than per-token routing for single-RGB SOD. To our knowledge no prior work combines spatial token-level routing, independent routers at several pyramid scales, and routing-informed decoding for single-RGB SOD without weather supervision. Sparse MoE has been studied in dense prediction – MoE-SPNet routes at the scene level [3], token-level routing has been applied to medical segmentation (SegMoTE [4]), and Pavlitska et al. examine the behaviour of sparse MoE layers in CNN segmentation [5] – and spatial MoE has been applied in a classification setting [6]. None of these addresses adverse-weather SOD with a single RGB input. This paper’s central finding is a null result: the spatial MoE we designed does not measurably affect predictions (Sections IV-F, IV-G). We report the architecture and its numbers together with the evidence that they are not attributable to routing.

Our contributions:

- 1) A token-level spatial router that assigns each token to top- $K=2$ experts from 8 per scale, using depthwise-convolutional context, noisy top- K gating, and true sparse dispatch.
- 2) Three independent routers at scales $1/4$, $1/8$, $1/16$, each with a separate expert pool.
- 3) Routing entropy over the full expert distribution, normalized by $\log 8$ for $E=8$, is injected as an explicit decoder feature via learnable additive fusion.
- 4) An ablation and inference-time intervention study (11 trained arms, 4 perturbations) showing that this routing mechanism is inert.

Code, configurations and the per-run evaluation artifacts are available at [7].

The rest of the paper is organised as follows. Section II positions the work against adverse-weather SOD and mixture-of-experts methods. Section III describes the architecture: the multi-scale backbone, the spatial router, sparse dispatch, the entropy-fused decoder and the training objective. Section IV reports the results — the headline numbers, a comparison against published methods, and a controlled ablation suite — and explains why the ablation behaves as it does. Section V states what the evidence supports and what it does not.

II. RELATED WORK

Two strategies address SOD under adverse weather. Weather-label-conditioned models such as NIFM [1] fuse noise indicators from explicit weather-type one-hot vectors, and WXSOD [8] supplies the benchmark and its own weather-aware baseline; both need weather supervision. Mixture-of-experts has instead been applied to adverse-weather restoration, notably WM-MoE [2], which routes with weather-aware gating and targets pixel regression rather than segmentation. Within SOD, existing MoE methods explore modality-, scale- or task-level routing for multi-modal inputs rather than per-token routing for single-RGB imagery. Token-level top- K routing was established by V-MoE [9] for classification, where routing entropy has also served for uncertainty estimation; we inject it as a decoder feature for dense prediction.

III. METHOD

Fig. 1 illustrates the full architecture. A shared PVTv2-B4 backbone extracts multi-scale features; three independent SpatialMoE layers process tokens at each scale; a cross-attention decoder fuses the results and produces dense predictions. We detail each component below.

A Multi-Scale Features. A PVTv2-B4 backbone [10] pre-trained on ImageNet extracts features at three scales via `out_indices = (0, 1, 2)`. Each scale is projected to $C=256$

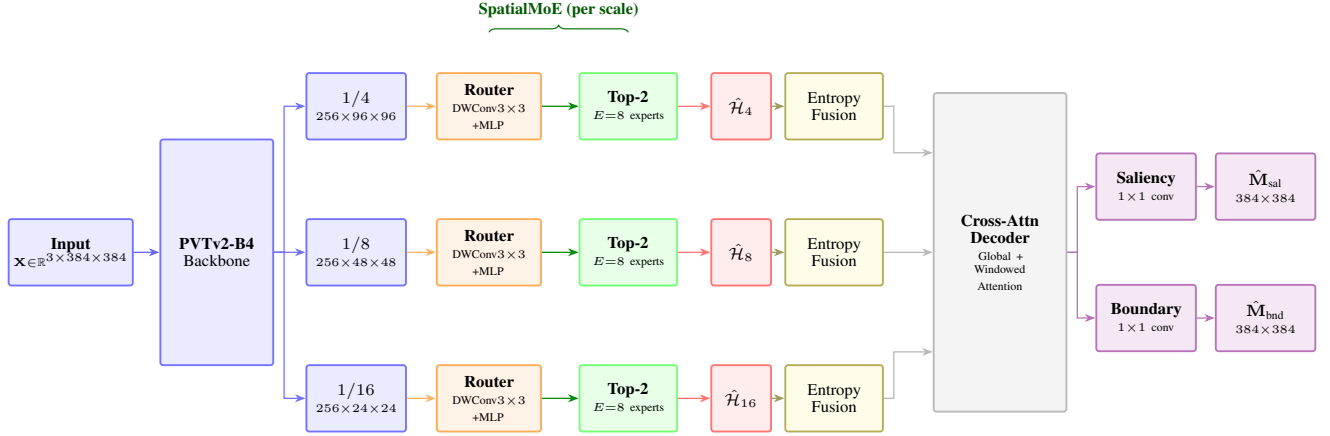


Figure 1. SpatialMoE-SOD architecture. Input $\mathbf{X} \in \mathbb{R}^{3 \times 384 \times 384}$ is processed by a PVTv2-B4 backbone to produce multi-scale features at 1/4, 1/8, 1/16 resolution. Each scale is independently processed by a SpatialMoE layer: a depthwise convolutional router selects top- $K=2$ experts from 8 via noisy gating, and a per-token entropy map $\hat{\mathcal{H}}$ is computed. The decoder merges scales via global and windowed cross-attention, with entropy features injected via learned additive fusion at each scale. Two 1×1 conv heads produce saliency and boundary logits at full resolution.

via 1×1 convolution:

$$\mathbf{X}_s = \text{Proj}_s(\text{Backbone}(\mathbf{X})_s) \in \mathbb{R}^{C \times H_s \times W_s}, \quad (1)$$

where $s \in \{4, 8, 16\}$ with $(H_s, W_s) \in \{(96, 96), (48, 48), (24, 24)\}$. Each feature map is reshaped into $N_s = H_s \times W_s$ tokens $\mathbf{x}_i \in \mathbb{R}^C$.

B Router. For each scale, a depthwise 3×3 convolution captures local spatial context from the feature map; the full routing mechanism is shown in Fig. 2. The router input concatenates each token with its local context:

$$\mathbf{r}_i = [\mathbf{x}_i \mid \text{DWConv}_{3 \times 3}(\mathbf{X}_s)_i] \in \mathbb{R}^{2C}, \quad (2)$$

where $\mid$ denotes channel-wise concatenation, yielding a $2C$ -dimensional vector that encodes both the token’s content and its spatial neighborhood. A two-layer MLP produces routing logits:

$$\mathbf{h}_i = W_2 \cdot \text{GELU}(W_1 \mathbf{r}_i + b_1) + b_2 \in \mathbb{R}^E, \quad (3)$$

with $W_1 \in \mathbb{R}^{128 \times 2C}$, $W_2 \in \mathbb{R}^{E \times 128}$, $E=8$ experts. The hidden dimension 128 provides a bottleneck between the $2C=512$ input and $E=8$ output.

C Noisy Top- K Routing. During training, Gaussian noise is added with input-dependent scale to encourage exploration:

$$\tilde{h}_{i,j} = h_{i,j} + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0, \sigma_j^2(\mathbf{x}_i)), \quad (4)$$

where $\sigma_j(\mathbf{x}_i) = \text{softplus}(w_j^\top \mathbf{x}_i + b_j)$ is a learned per-expert, per-token noise scale. At inference, noise is disabled ($\epsilon=0$). Top- $K=2$ experts are selected per token; gate weights are softmax over selected logits only:

$$g_{i,k} = \frac{\exp(\tilde{h}_{i,k})}{\sum_{j \in \mathcal{T}_i} \exp(\tilde{h}_{i,j})}, \quad k \in \mathcal{T}_i = \text{top-k}(\tilde{h}_{i,j}, K=2). \quad (5)$$

This produces a sparse assignment: each token attends to exactly 2 of 8 experts, and each expert processes only the tokens that selected it.

D Sparse Expert Aggregation. Each expert $\mathcal{E}_k$ executes only on tokens that selected it. For expert k , let $\mathcal{A}_k = \{i : k \in \mathcal{T}_i\}$ denote its assigned tokens:

$$\mathbf{y}_i = \sum_{k \in \mathcal{T}_i} g_{i,k} \cdot \mathcal{E}_k(\mathbf{x}_i), \quad (6)$$

implemented via a loop over experts with mask-based selection (true sparsity, not dense with masking). Each expert is a token-wise residual MLP:

$$\mathcal{E}_k(\mathbf{x}) = \mathbf{x} + W_k^{(2)} \cdot \text{GELU}(W_k^{(1)} \cdot \text{LN}(\mathbf{x})), \quad (7)$$

with $W_k^{(1)} \in \mathbb{R}^{4C \times C}$, $W_k^{(2)} \in \mathbb{R}^{C \times 4C}$ ($4 \times$ expansion from $C=256$ to $4C=1024$). Each scale has 8 independent experts with no weight sharing (24 total). Reshaping per-token outputs yields $\mathbf{Y}_s \in \mathbb{R}^{C \times H_s \times W_s}$.

E Routing Entropy. Per-token routing entropy measures assignment uncertainty over the full expert distribution:

$$\mathcal{H}_i = - \sum_{j=1}^E p_{i,j} \log p_{i,j}, \quad \hat{\mathcal{H}}_i = \mathcal{H}_i / \log 8 \in [0, 1], \quad (8)$$

where $p_{i,j}$ is the router’s softmax probability for expert j — computed over all E experts, not only the selected ones. The normalizing constant is hard-coded to $\log 8$, matching the eight-expert model, so $\hat{\mathcal{H}}_i \in [0, 1]$: 0 indicates a single expert dominates and 1 a uniform assignment over all eight experts. The normalized entropy map $\hat{\mathcal{H}}_s \in \mathbb{R}^{1 \times H_s \times W_s}$ is fused with MoE output via a learned additive block:

$$\mathbf{F}_s = \text{Proj}_Y(\mathbf{Y}_s) + \lambda_s \cdot \text{Proj}_{\mathcal{H}}(\hat{\mathcal{H}}_s), \quad (9)$$

where $\text{Proj}_{\mathcal{H}}$ is a 1×1 convolution from 1 to C channels and λ_s is a learnable scalar initialized to 0.1. This allows the decoder to condition its attention on routing confidence without requiring the entropy to be differentiable through the expert weights.

F Decoder. The decoder merges coarse-to-fine features via top-down cross-attention. At the coarsest scale, $\mathbf{F}_{16}$ is upsampled and used as queries with $\mathbf{F}_8$ as keys and values (8-head global attention):

$$\mathbf{F}'_8 = \text{GlobalCrossAttn}(Q = \text{Up}(\mathbf{F}_{16}), KV = \mathbf{F}_8). \quad (10)$$

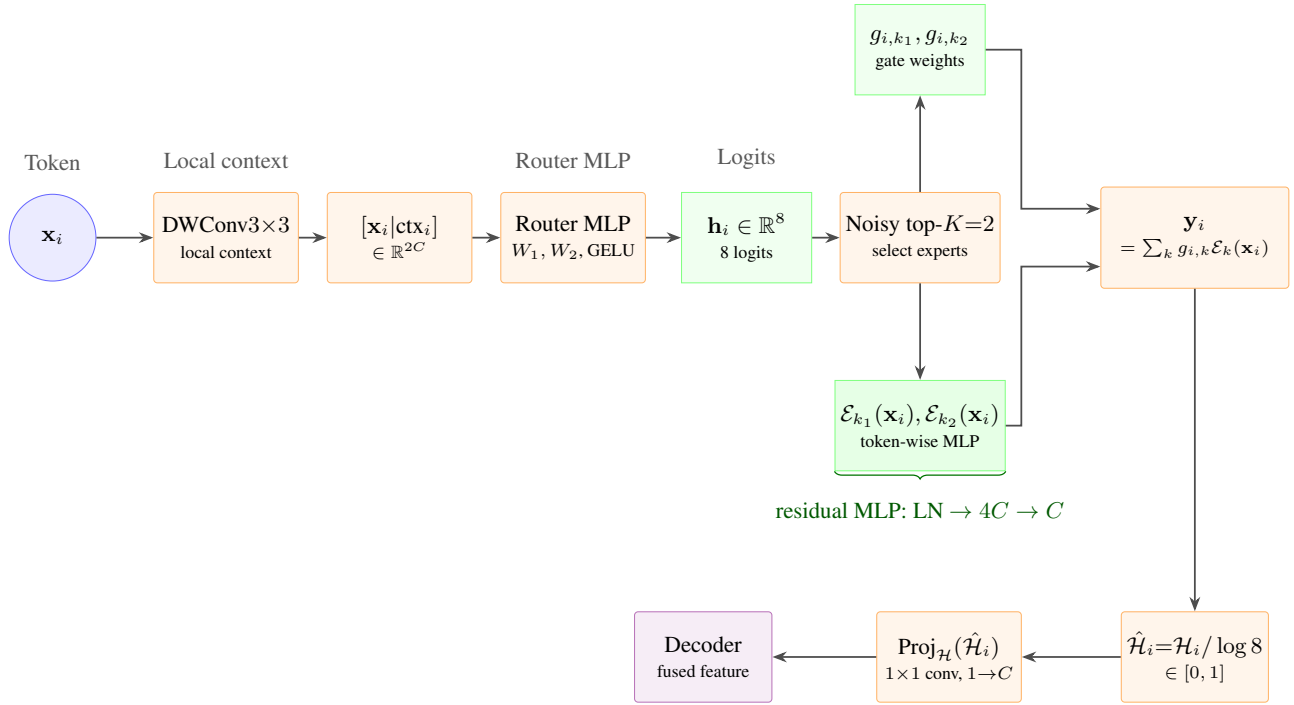


Figure 2. Routing mechanism for token $\mathbf{x}_i$ at one pyramid scale. A depthwise 3×3 convolution extracts local spatial context; the concatenation $\mathbf{r}_i = [\mathbf{x}_i | \text{ctx}_i] \in \mathbb{R}^{2C}$ is passed through a two-layer MLP to produce 8 routing logits. Noisy top- $K=2$ selection (Gaussian noise during training) picks two experts, branching into gate weights (top) and expert outputs (bottom); gate weights are softmax over selected logits only. The selected experts process the token independently; their outputs are summed with gate weights. Routing entropy $\mathcal{H}_i \in [0, 1]$ measures assignment uncertainty and is injected as a feature channel into the decoder via a learned 1×1 projection.

Global context is then injected into the fine scale: $\mathbf{F}_4^{\text{ctx}} = \mathbf{F}_4 + \text{Proj}_g(\text{AvgPool}(\mathbf{F}_8'))$. $\mathbf{F}_8'$ is upsampled and used as queries with $\mathbf{F}_4^{\text{ctx}}$ as keys and values using 7×7 windowed attention for efficiency:

$$\mathbf{F}_4' = \text{WinCrossAttn}(Q = \text{Up}(\mathbf{F}_8'), KV = \mathbf{F}_4^{\text{ctx}}). \quad (11)$$

Three refinement blocks (each: Conv $3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{Conv } 3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{residual addition}$) progressively upsample from $1/4$ to full resolution. Two 1×1 convolution heads produce saliency logits $\hat{\mathbf{M}}_{\text{sal}}$ and boundary logits $\hat{\mathbf{M}}_{\text{bnd}}$ at 384×384 .

G Training Objective. The objective is a weighted sum of the eight terms that are active in the reported configuration (the Z-loss and the router-confidence term are disabled):

$$\mathcal{L} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{IoU}} \mathcal{L}_{\text{IoU}} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \lambda_{\text{BND}} \mathcal{L}_{\text{BND}} + \lambda_{\text{LB}} \mathcal{L}_{\text{LB}} + \lambda_{\text{IMP}} \mathcal{L}_{\text{IMP}} + \lambda_{\text{auxBND}} \mathcal{L}_{\text{auxBND}} + \lambda_{\text{DS}} \mathcal{L}_{\text{DS}}, \quad (12)$$

with $\lambda_{\text{BCE}} = \lambda_{\text{IoU}} = \lambda_{\text{SSIM}} = \lambda_{\text{BND}} = 1.0$, $\lambda_{\text{LB}} = \lambda_{\text{IMP}} = 0.01$, $\lambda_{\text{auxBND}} = 0.5$ and $\lambda_{\text{DS}} = 0.4$ (λ_{LB} is applied at each of the three scales). $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy with logits between predicted saliency logits and ground-truth mask. $\mathcal{L}_{\text{IoU}}$ is the soft IoU loss: $1 - \text{mean}_b \left[\frac{\sum_{hw} \sigma(\hat{M}_b) \cdot M_b + \epsilon}{\sum_{hw} \sigma(\hat{M}_b) + \sum_{hw} M_b - \sum_{hw} \sigma(\hat{M}_b) \cdot M_b + \epsilon} \right]$. $\mathcal{L}_{\text{SSIM}}$ is 1 minus the structural-similarity index between the sigmoid prediction and the mask. $\mathcal{L}_{\text{BND}}$ is a smooth- L_1 term between the magnitude of the predicted spatial gradient and the ground-truth boundary map. $\mathcal{L}_{\text{LB}}$ is the load-balancing loss: $\sum_s E \sum_j f_j^{(s)} P_j^{(s)}$, summed over the three scales, where f_j is the fraction of all routing slots assigned to expert j , computed from hard assignments and detached, and P_j is the

differentiable mean gate mass. $\mathcal{L}_{\text{IMP}}$ is the importance loss: $\frac{1}{|\mathcal{S}|} \sum_s (\text{std}(\mathbf{P}^{(s)}) / \text{mean}(\mathbf{P}^{(s)}))^2$, where $\mathbf{P}^{(s)} \in \mathbb{R}^E$ denotes the per-expert mean gate-mass vector at scale s . $\mathcal{L}_{\text{auxBND}}$ is binary cross-entropy between the boundary head's logits and the boundary map. $\mathcal{L}_{\text{DS}}$ is the mean binary cross-entropy of the three per-scale auxiliary saliency heads; the auxiliary heads exist because deep supervision is enabled in the reported configuration.

IV. EXPERIMENTS

A Setup. We evaluate on WXSOD [8], which provides 12,891 training images (scene-aware 80/20 split), 1,500 synthetic test images across 9 weather categories, and 554 real-world test images across 5 categories (fog, rain, snow, dark, low-light). Images are aspect-preserving resized to 384×384 ; augmentations include horizontal flip ($p=0.5$), color jitter ($p=0.5$), and ImageNet normalization. Training uses DDP on 2 GPUs, AMP (FP16), AdamW optimizer (lr 10^{-4} for both newly introduced modules and the backbone, weight decay 10^{-4}), WarmupCosine schedule (1% warmup, 50 epochs), gradient clipping at 1.0, effective batch size 32. The backbone is frozen during the first epoch and unfrozen from epoch 1 onward. Metrics: MAE, S_ϕ [11], E_ϕ^{adp} [12], F_β^{max} [13]. The model has 69.21M parameters and 277.9G MACs per 384×384 input image.

What we find. Four results, each backed by a table below.

- **The mixture-of-experts does not change the prediction.** Removing it entirely leaves real-test MAE at 0.0195, identical to the full model, and routing at one pyramid scale instead of three gives the same 0.0195 (Table 5).
- **Competitive without weather labels.** MAE 0.0192 on the 1,500 synthesized test images leads every method in the published benchmark, including the weather-supervised WFANet

(0.0229); 0.0168 on the 554 real images is second of nineteen (Table 3).

- **The router is not making a meaningful choice.** Routing entropy sits at 99.8–100% of its log E ceiling in every arm, and expert weights are orthogonal while their activations correlate at 0.64 (Tables 11 and 13).
- **Training budget matters more than any design choice.** The same recipe at 14 epochs instead of 8 improves real-test MAE from 0.0195 to 0.0174, against a seed spread of ± 0.0004 (Tables 5 and 14).

B Main Results. Table 1 reports global performance on both test splits. The model achieves MAE of 0.0192 on synthetic data and 0.0168 on real-world data. Real-world MAE (0.0168) is slightly lower than synthetic MAE (0.0192). These numbers are driven primarily by the backbone and the training schedule rather than by the routing mechanism under study (Section IV-E). However, this comparison is confounded by test-set composition: the synthetic test set includes compound weather conditions (rain+fog, rain+snow, snow+fog) that are absent from the real-world test set, which may increase synthetic-test error independently of any domain-shift effect.

Table 1. Global performance on WXSOD test splits.

Set	MAE $\downarrow$	S_ϕ $\uparrow$	$F_\beta^{\max}$ $\uparrow$	E_ϕ^{adp} $\uparrow$
Synth.	0.0192	0.9139	0.9015	0.9591
Real	0.0168	0.9151	0.8936	0.9530

C Extended Metrics. Table 2 reports the remaining metrics for the reported model, and Tables 6 and 7 gives the same six metrics for every arm in the ablation table. All three E-measure and F-measure variants agree with the trends above. The boundary measures are the weakest component: boundary F1 is 0.3601 on real test. This is expected: the reported configuration trains with the boundary term inactive, and given that routing is itself inert, boundary supervision is the more promising direction for architectural work.

Table 2. Extended metrics for the reported model. Real split $N=554$, synthetic $N=1500$.

Metric	Synthetic	Real
E_ϕ^{mean}	0.9585	0.9551
$E_\phi^{\max}$	0.9633	0.9608
F_β^{mean}	0.8886	0.8747
$F_\beta^{\max}$	0.9016	0.8936
Boundary MAE	0.4817	0.6334
Boundary F1	0.5472	0.3601

D Comparison With Published Methods. WXSOD publishes results for 17 SOD methods and its own weather-supervised baseline WFANet on exactly these two test splits, under the same metric family [8]. Table 3 places our model alongside them. On the synthesized split our model attains the lowest MAE in that table, including WFANet, which is trained with explicit weather-class labels. On the real split it is second of nineteen, behind WFANet and ahead of every other method. The published numbers were produced by their authors under their own protocol; ours are measured here, so this is a like-for-like comparison in dataset, splits and metrics rather than a re-implementation, and it is labelled as reported. Our model is also larger than WFANet (69.21M against 50.87M parameters), which we state plainly rather than present as an efficiency claim.

Table 3. Comparison with published WXSOD results. Published values are as reported in [8] (Tables II and III); ours are measured here. WFANet uses weather class labels, our model uses none. Lower MAE and higher S_ϕ are better.

Method	Synthetic		Real	
	MAE $\downarrow$	S_ϕ $\uparrow$	MAE $\downarrow$	S_ϕ $\uparrow$
MINet [14]	0.0775	0.7403	0.0549	0.7710
ADMNet [15]	0.0770	0.7497	0.0544	0.7839
TRACER [16]	0.0280	0.8898	0.0227	0.8961
SEANet [17]	0.0332	0.8694	0.0239	0.8978
A3Net [18]	0.0327	0.8784	0.0263	0.8909
DCNet [19]	0.0320	0.8792	0.0215	0.9058
MSRMNet [20]	0.0275	0.8895	0.0198	0.9091
SAFINet [21]	0.0384	0.8643	0.0217	0.9050
TCGNet [22]	0.0317	0.8832	0.0185	0.9185
GeleNet [23]	0.0239	0.9038	0.0192	0.9186
GPONet [24]	0.0266	0.8962	0.0188	0.9098
WFANet [8] (<i>weather-supervised</i>)	0.0229	0.9051	0.0159	0.9248
Ours (no weather labels)	0.0192	0.9140	0.0168	0.9151

E Ablations. Table 5 reports every arm we trained. The middle block shares one recipe — renormalized gate, router noise enabled, effective batch 32, 8 epochs — and each row changes a single field, so those rows are directly comparable to one another.

Three findings stand out. First, **removing the mixture-of-experts entirely leaves performance unchanged:** the no-MoE arm scores 0.0195 on real test against 0.0195 for the full model, with S_ϕ agreeing to 0.0010. Second, **reducing routing from three pyramid scales to one changes nothing** (0.0195), and replacing the routed experts with a single dense expert per scale changes nothing (0.0191). Third, the only architectural knob that moves the metric is top- k : routing to a single expert costs 0.0017 MAE (0.0212 against 0.0195), roughly four times the seed spread of ± 0.0004 measured across seeds 42–44.

(The A2, A4 and A5 deltas of 0.0002, 0.0001 and 0.0004 sit inside that band and are not meaningful; only the top- k result at 0.0017 and the epoch count at 0.0021 exceed it.)

That floor is measured on the reference configuration, and the remaining arms are single-seed, so their deltas are compared against a floor estimated elsewhere. A paired test over per-image errors would settle this directly and is not reported here.

Training budget is the only variable that matters more than any architectural choice: the same recipe and seed trained for 14 epochs instead of 8 improves real-test MAE from 0.0195 to 0.0174, a larger effect than every design-choice difference we measured combined. We therefore report the architecture as characterisation, not as a demonstrated improvement.

Table 4 defines a short code for every trained arm; Table 5 and every subsequent per-arm table (Tables 6–13) refer to arms by this code alone, so the full configuration is stated once rather than repeated verbatim in five different tables.

F Inference-Time Ablations. Table 8 perturbs the *trained* reference model at inference only, so it isolates how much each component contributes to the final prediction rather than what training does. Replacing the learned router with **random routing** leaves real-test MAE within 0.0005 of the full model. Disabling entropy fusion changes it by 0.0001, and forcing every token at the 1/4 scale to a single expert changes it by -0.0001 . None of the mechanism’s parts affects the output: the mixture, its routing and its entropy channel are all inert at inference.

G Why Routing Does Not Matter. Why does none of it matter? Two measurements answer this, and together they are the mechanistic contribution of the paper.

The router is indifferent. Routing entropy sits at 99.8–100%

Table 4. Arm codes used in Tables 5, 6, 6, 13 and 13. **R** is the reported model; **Ref** is the ablation-family baseline; **A1–A10** change one field from **Ref**; **B1–B2** use an earlier, non-comparable recipe (batch 40).

Code	Configuration
<i>Reported model</i> (14 epochs, earlier recipe)	
R	E8 $k=2$
<i>Ablation family</i> (one recipe; single field changed per row)	
Ref	E8 $k=2$ (reference, 8 epochs)
A1	E8 $k=2$, 14 epochs
A2	E2 $k=2$
A3	E4 $k=1$
A4	E4 $k=2$, load-balance weight 0
A5	E4 $k=2$, dense expert type
A6	E4 $k=2$, no MoE at all
A7	E8 $k=2$, dense gate
A8	E8 $k=2$, routing at 1/4 only
A9	E8 $k=2$, seed 43
A10	E8 $k=2$, seed 44
<i>Earlier recipe generation</i> (batch 40)	
B1	E2 $k=1$, dense gate
B2	E4 $k=2$, dense gate

Table 5. All trained arms (codes per Table 4). Real split $N=554$, synthetic $N=1500$.

Arm	Real		Synthetic	
	MAE↓	S_ϕ ↑	MAE↓	S_ϕ ↑
R	0.0168	0.9151	0.0192	0.9140
Ref	0.0195	0.9076	0.0214	0.9085
A1	0.0174	0.9169	0.0202	0.9141
A2	0.0193	0.9087	0.0210	0.9088
A3	0.0212	0.8985	0.0231	0.9036
A4	0.0196	0.9067	0.0212	0.9083
A5	0.0191	0.9086	0.0212	0.9080
A6	0.0195	0.9066	0.0210	0.9084
A7	0.0199	0.9060	0.0212	0.9084
A8	0.0195	0.9077	0.0212	0.9078
A9	0.0199	0.9048	0.0210	0.9086
A10	0.0191	0.9091	0.0213	0.9083
B1	0.0199	0.9034	0.0215	0.9041
B2	0.0197	0.9043	0.0215	0.9040

Table 6. Extended metrics for every arm, real test ($N=554$). bMAE and bF1 are boundary MAE and boundary F1; the ablation arm labels follow Table 5.

Arm	E^{mean}	E^{max}	F^{mean}	F^{max}	bMAE	bF1
R	0.9551	0.9608	0.8747	0.8936	0.6334	0.3601
Ref	0.9467	0.9567	0.8543	0.8854	0.7211	0.2876
A1	0.9554	0.9631	0.8719	0.8963	0.6675	0.3373
A2	0.9491	0.9583	0.8584	0.8860	0.7186	0.2912
A3	0.9376	0.9560	0.8293	0.8801	0.7688	0.2390
A4	0.9471	0.9571	0.8534	0.8860	0.7260	0.2835
A5	0.9492	0.9581	0.8583	0.8868	0.7174	0.2915
A6	0.9479	0.9581	0.8541	0.8841	0.7250	0.2835
A7	0.9460	0.9566	0.8525	0.8854	0.7317	0.2809
A8	0.9495	0.9592	0.8584	0.8869	0.7221	0.2881
A9	0.9452	0.9552	0.8498	0.8822	0.7345	0.2747
A10	0.9478	0.9583	0.8565	0.8854	0.7157	0.2922
B1	0.9469	0.9546	0.8535	0.8794	0.7345	0.2739
B2	0.9483	0.9559	0.8561	0.8804	0.7330	0.2768

Table 7. Extended metrics for every arm, synthetic test ($N=1,500$). bMAE and bF1 are boundary MAE and boundary F1; the ablation arm labels follow Table 5.

Arm	E^{mean}	E^{max}	F^{mean}	F^{max}	bMAE	bF1
R	0.9585	0.9633	0.8886	0.9016	0.4817	0.5472
Ref	0.9516	0.9617	0.8715	0.8925	0.5534	0.4947
A1	0.9536	0.9617	0.8812	0.8975	0.5067	0.5348
A2	0.9519	0.9611	0.8722	0.8922	0.5521	0.4968
A3	0.9440	0.9592	0.8514	0.8878	0.5874	0.4577
A4	0.9511	0.9604	0.8706	0.8917	0.5544	0.4941
A5	0.9512	0.9602	0.8715	0.8902	0.5540	0.4941
A6	0.9521	0.9615	0.8708	0.8909	0.5557	0.4907
A7	0.9508	0.9608	0.8692	0.8917	0.5612	0.4895
A8	0.9503	0.9588	0.8705	0.8889	0.5554	0.4929
A9	0.9514	0.9609	0.8688	0.8920	0.5559	0.4914
A10	0.9516	0.9604	0.8714	0.8911	0.5525	0.4945
B1	0.9516	0.9588	0.8687	0.8862	0.5676	0.4782
B2	0.9522	0.9592	0.8692	0.8862	0.5685	0.4771

Table 8. Inference-time ablations on the reference arm (real test, $N=554$). Random routing is the mean over 4 seeds.

Inference variant	MAE↓	S_ϕ ↑
Full model	0.0197	0.9043
Random routing (mean of 4)	0.0193	0.9060
Entropy fusion disabled	0.0197	0.9044
Force scale 1/4 to one expert	0.0196	0.9048

of its log E ceiling (Table 11), so the distribution over experts is essentially uniform: the router almost never expresses a preference.

The experts are redundant. Table 9 compares expert pairs at each scale. Pairwise cosine similarity between expert *weights* is essentially zero, so the experts are not copies of one another. But cosine similarity between their *activations* on real data is 0.6424 at scale 1/4, 0.2733 at 1/8 and 0.8617 at 1/16. Despite different weights, the experts’ outputs are substantially correlated at two of the three scales, so for most tokens which expert they are routed to changes little.

Fig. 3 shows the same insensitivity qualitatively, and Table 13 extends the similarity measurement to every arm. This also explains why the one knob that matters is top- k : with $k=2$ the output averages two correlated experts, suppressing their individual idiosyncrasies, whereas $k=1$ takes a single expert’s output and is measurably worse (0.0212 against 0.0195).

Table 9. Routing diagnostics on the reference arm at the final logged epoch. Usage is the hard assignment fraction per expert (uniform would be 12.5% for $E=8$); dead marks experts below 2%. Similarity is the mean pairwise cosine over expert pairs, on real test.

Scale	Usage % per expert	Dead	Wt. sim.	Act. sim.
1/4	21.1, 6.3, 5.3, 5.6, 1.5, 23.8, 29.7, 6.7	[4]	+0.0000	0.6424
1/8	1.1, 6.9, 3.1, 4.2, 15.5, 0.8, 40.7, 27.7	[0, 5]	+0.0003	0.2733
1/16	9.2, 5.1, 14.1, 13.1, 4.1, 27.1, 16.3, 11.1	—	+0.0010	0.8617

H Weather-Wise Analysis. Table 10 reports per-weather performance on real-world test, and Fig. 4 plots it. Performance varies by $2.6\times$ across weather types: snow yields the lowest MAE (0.0094, $N=90$), while low-light yields the highest (0.0243, $N=93$). Fog (0.0148, $N=126$), rain (0.0165, $N=120$), and dark (0.0191, $N=125$) fall between these extremes. On synthetic data, compound weather conditions (rain+fog) yield the worst MAE (0.0237), suggesting that compositing multiple degradation types compounds difficulty.

Table 10. Weather-wise results on real-world test (554 images).

Weather	N	MAE ↓	S_ϕ ↑	F_β^{max} ↑
Snow	90	0.0094	0.9478	0.9421
Fog	126	0.0148	0.9179	0.8914
Rain	120	0.0165	0.9157	0.8935
Dark	125	0.0191	0.9072	0.8885
Low-light	93	0.0243	0.8897	0.8629

I Routing Entropy. Table 11 reports the mean routing entropy per scale. Entropy is taken over the full softmax over all $E=8$ experts, so the ceiling is $\log 8 = 2.0794$ nats; we therefore report both the raw value and its fraction of that maximum. The measured mean sits within **0.2%** of the ceiling at every scale on both test splits. In other words the router’s distribution over the eight experts is, on average, very nearly uniform: it expresses almost no preference between them. This is a statement about the router’s decisiveness, not about its accuracy—the routing is close to uniform, so the per-token choice of expert carries little information.

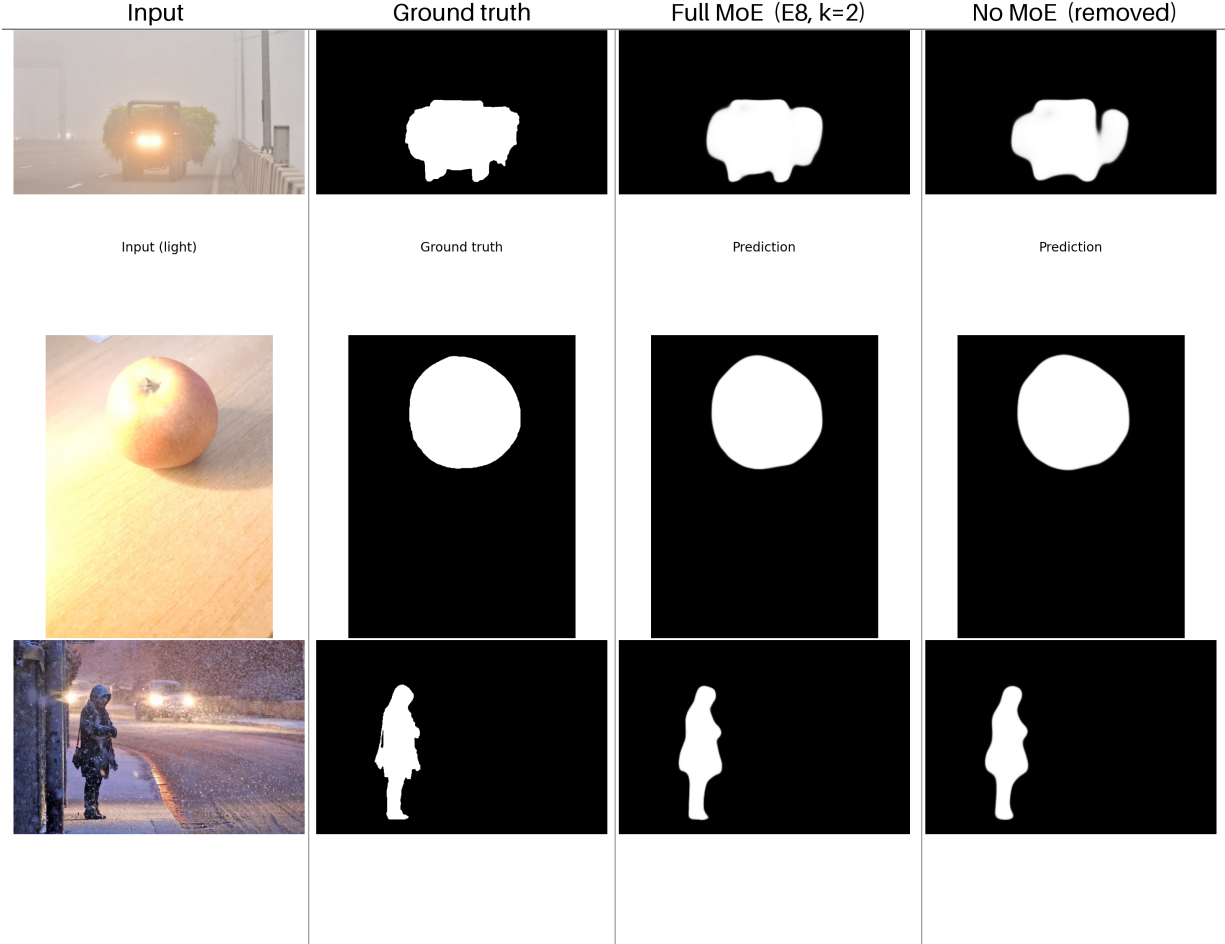


Figure 3. The mixture-of-experts does not change the prediction. Read the four columns left to right: (1) input, (2) ground truth, (3) full MoE (E_8 , $k=2$, 14 epochs) and (4) no MoE (the identical architecture with the mixture removed). Columns 3 and 4 are therefore two different models on the same image; both are labelled “Prediction” inside the grid because that is how the evaluation tool renders a single model’s output. Rows are real-world test images under fog, low-light and snow. The two prediction columns are indistinguishable in the low-light row; in the fog and snow rows the differences are confined to object boundaries. This is the qualitative counterpart of the no-MoE row in Table 5, where removing the MoE leaves real-test MAE unchanged at 0.0195.

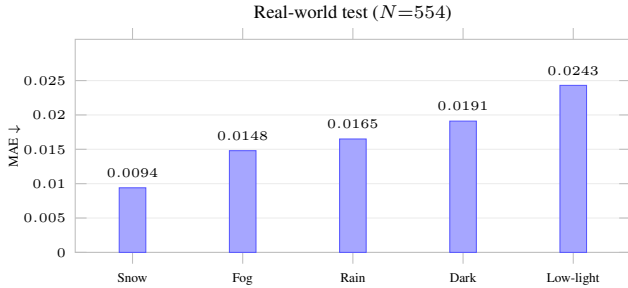


Figure 4. Weather-wise MAE on real-world test. Snow yields the lowest error, while low-light yields the highest.

Table 11. Mean routing entropy per scale. Entropy is over the full distribution over all $E=8$ experts, so the maximum is $\log 8 = 2.0794$ nats.

Scale	Synthetic		Real	
	nats	% of max	nats	% of max
1/4	2.0786	99.96	2.0788	99.97
1/8	2.0788	99.97	2.0789	99.97
1/16	2.0755	99.81	2.0756	99.82

J Weather-Wise Analysis: Synthetic. The synthetic test set spans nine conditions, including three compound ones. Table 12 reports them, ordered worst first. The compound conditions are the

hardest, and the clean subset is not the easiest — several single-degradation conditions are harder than the clean images, so the difficulty ordering is not simply a function of how much noise was injected.

Table 12. Weather-wise results on synthetic test (1,500 images), ordered by MAE.

Condition	N	MAE↓	S_ϕ ↑	E_ϕ^{adp} ↑	F_β^{max} ↑
rainafog	172	0.0237	0.8903	0.9384	0.8691
light	166	0.0215	0.9091	0.9623	0.9046
rain	179	0.0200	0.9173	0.9542	0.9030
dark	156	0.0199	0.9049	0.9539	0.8836
rainasnow	166	0.0194	0.9152	0.9637	0.9032
snowafog	156	0.0189	0.9051	0.9567	0.8950
snow	172	0.0189	0.9192	0.9647	0.9068
fog	166	0.0166	0.9308	0.9696	0.9255
clean	167	0.0142	0.9329	0.9690	0.9229

K Routing Diagnostics for Every Arm. Table 13 reports, for every trained arm, the normalised routing entropy at each scale and the number of dead experts (usage below 2%). The uniformity is not a property of one configuration: every arm sits at essentially 100% of its $\log E$ ceiling, including the arms that remove routing entirely at two scales and the arm that removes the load-balancing loss that is supposed to enforce balance.

L Expert Similarity for Every Arm. Table 13 reports mean pair-wise cosine similarity between expert weights and between their

Table 13. Diagnostics for every arm. Left: routing entropy per scale, normalised by $\log E$ (1.000 = a uniform distribution over the eight experts), and the number of dead experts (usage below 2%). Right: mean pairwise cosine similarity between experts, over weight vectors and over activations, real test. R is the reported checkpoint, measured by its own diagnostic pass; the rest are the ablation arms. “—” means not produced for that arm; “n/a” means the quantity is undefined because the scale has fewer than two experts (the pass-through scales of A8, and the single shared expert of A5, A6).

Arm	Routing entropy $H/\log E$ (dead experts)						Expert similarity					
	1/4	dead	1/8	dead	1/16	dead	wt 1/4	act 1/4	wt 1/8	act 1/8	wt 1/16	act 1/16
R (<i>reported</i>)	0.9997	0	0.9997	2	0.9982	0	-0.0000	0.6383	+0.0002	0.2731	+0.0011	0.8748
Ref	0.9996	1	0.9997	2	0.9980	0	+0.0000	0.6424	+0.0003	0.2733	+0.0010	0.8617
A1	0.9996	1	0.9998	2	0.9972	0	+0.0000	0.6584	+0.0003	0.2621	+0.0011	0.8830
A2	0.9994	0	0.9993	0	0.9923	0	-0.0002	0.7124	-0.0007	0.4914	+0.0007	0.8570
A3	0.9983	1	0.9766	3	0.9933	3	+0.0007	0.5687	-0.0003	0.0739	+0.0011	0.8671
A4	0.9996	0	0.9998	0	0.9969	0	+0.0007	0.6561	-0.0002	0.2998	+0.0010	0.8556
A5	0.0000	3	0.0000	3	0.0000	3	—	—	—	—	—	—
A6	0.0000	3	0.0000	3	0.0000	3	—	—	—	—	—	—
A7	1.0000	0	1.0000	0	0.9996	0	+0.0000	0.7013	+0.0002	0.5607	+0.0015	0.9438
A8	0.9996	1	0.0000	7	0.0000	7	+0.0000	0.6424	—	—	—	—
A9	0.9995	1	0.9998	0	0.9972	1	-0.0005	0.6472	-0.0004	0.2631	+0.0002	0.8622
A10	0.9995	1	0.9997	1	0.9973	0	+0.0001	0.6412	-0.0001	0.2790	+0.0006	0.8614
B1	1.0000	0	1.0000	0	0.9979	0	-0.0002	0.6413	-0.0008	0.0795	+0.0008	0.8908
B2	1.0000	0	1.0000	0	0.9974	0	+0.0006	0.6351	-0.0002	0.4469	+0.0011	0.8669

activations. Across every arm, expert *weights* are essentially orthogonal, while expert *activations* are positively correlated — the experts are trained to different parameters that nevertheless respond similarly to real weather images.

M Seed Variance. Table 14 repeats one configuration under three seeds. Real-test MAE spans 0.0008 (standard deviation 0.0003) and synthetic MAE spans 0.0004. This is the noise floor for every comparison in this paper: differences smaller than about 0.0005 MAE are not interpretable.

Table 14. Seed variance for E8 $k=2$, 8 epochs, everything else fixed.

Seed	Real			Synthetic		
	MAE	S_ϕ	E^{adp}	MAE	S_ϕ	E^{adp}
42	0.0195	0.9076	0.9402	0.0214	0.9085	0.9493
43	0.0199	0.9048	0.9380	0.0210	0.9086	0.9481
44	0.0191	0.9091	0.9412	0.0213	0.9083	0.9513

N Discussion. The model demonstrates that spatial token-level routing is feasible for single-RGB SOD without weather-specific supervision. However, several important limitations must be noted. First, the comparison against prior methods uses the benchmark’s published numbers rather than our own re-implementations, so it is a comparison as reported; WFANet still attains the best real-split MAE, so we do not claim state-of-the-art performance. Second, the reported model attains boundary F1 of only 0.3601 on real test. That follows from training with the boundary term inactive, and it is a limitation of the reported model rather than of the ablation. Third, the model was trained once without replicate runs, so variance estimates are unavailable. Third, the forced-expert analysis suggests that expert specialization may be limited, and the benefit of multi-expert routing over a single larger expert remains unclear. First and most centrally, the spatial MoE is shown across eleven trained arms and four inference-time interventions to have no measurable effect on output quality; we treat that as the paper’s primary contribution rather than a caveat. Routing entropy also sits at 99.8–100% of its ceiling, so the decoder’s entropy channel is nearly constant.

V. CONCLUSION

We presented and rigorously ablated SpatialMoE-SOD. The full model is competitive with weather-supervised baselines on WXSOD, but a complete ablation and intervention study shows the mixture-of-experts is not responsible for that performance: it can be removed, reduced to one scale, or replaced with a dense expert with no measurable effect – generalising to dense prediction a caution already familiar in the MoE literature. Per-token

routing entropy is normalized and injected as an explicit decoder feature via learnable additive fusion. On WXSOD, the model achieves MAE 0.0192 (synthetic, 1,500 images) and 0.0168 (real-world, 554 images), with weather-wise MAE ranging from 0.0094 (snow) to 0.0243 (low-light).

Several limitations warrant future investigation. The model was trained once without replicate runs, so variance estimates are unavailable. Against the benchmark’s published results the model leads the synthesized split and places second of nineteen on the real split, behind the weather-supervised WFANet [8], while using no weather labels. The forced-expert analysis suggests limited differentiation among experts at scale 1/4, and the contribution of the entropy fusion mechanism relative to a baseline without it has not been ablated. Routing entropy sits at 99.8–100% of the maximum attainable over the eight experts, so routing is close to uniform and the entropy channel carries little signal. The ablation matrix is otherwise complete: removing the mixture-of-experts, reducing routing to one pyramid scale, replacing routed experts with a dense expert and removing the load-balancing loss all leave real-test MAE within 0.0004 of the reference, while routing to a single expert costs 0.0017. The mixture is therefore not earning its parameters on this task, and the useful direction is not more routing capacity but the training budget. Future work should determine whether a routing objective that is actually supervised – rather than one optimised only through the saliency loss – can induce specialisation, and should re-run the benchmark’s baselines under a single unified protocol.

REFERENCES

- [1] Q. Chen, X. Yang, T. Wang, R. Lu, X. Sheng, Y. Sun, and B. Zheng, “Salient object detection in complex weather conditions via noise indicators,” *arXiv preprint arXiv:2512.10592*, 2025.
- [2] Y. Luo, R. Zhao, X. Wei, J. Chen, Y. Lu, S. Xie, T. Wang, R. Xiong, M. Lu, and S. Zhang, “WM-MoE: Weather-aware multi-scale mixture-of-experts for blind adverse weather removal,” *arXiv preprint arXiv:2303.13739*, 2023.
- [3] Z. Zhao, X. Qi, Y. Xu, *et al.*, “MoE-SPNet: A mixture-of-experts scene parsing network,” *Pattern Recognition*, 2018. arXiv:1806.07049.
- [4] Y. Lu, J. Li, S. Ju, *et al.*, “SegMoTE: Token-level mixture of experts for medical image segmentation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 36332–36342, 2026.
- [5] S. Pavlitska, H. Fan, K. Ditschuneit, and J. M. Zöllner, “Design and behavior of sparse mixture-of-experts layers in CNN-based semantic segmentation,” in *CVPR Workshops (Safe Artificial Intelligence for Automated Driving)*, pp. 4272–4281, 2026. arXiv:2604.13761.
- [6] N. Dryden, T. Hoefler, *et al.*, “Spatial mixture-of-experts,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2211.13491.
- [7] B. A. Reddy, “SpatialMoE-SOD: Source code, configurations and per-run evaluation artifacts.” <https://github.com/AviBapatu/Spatial-MoE-for-Adverse-Weather-Salient-Object-Detection>, 2026.
- [8] Q. Chen, X. Yang, R. Lu, Q. Zhang, Y. Liu, X. Zhou, and B. Zheng, “WXSOD: A benchmark for robust salient object detection in adverse weather conditions,” *Pattern Recognition*, vol. 170, p. 114003, 2026. Dataset repository: <https://github.com/C-water/WXSOD>.
- [9] C. Riquelme, J. Puigcerver, B. Mustafa, M. Neumann, R. Jenatton, A. Susano Pinto, D. Keysers, and N. Houlsby, “Scaling vision with sparse mixture of experts,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.
- [10] W. Wang, E. Xie, X. Li, D.-P. Fan, K. Song, D. Liang, T. Lu, P. Luo, and L. Shao, “PVT v2: Improved baselines with pyramid vision transformer,” *Computational Visual Media*, vol. 8, no. 3, pp. 415–424, 2022.
- [11] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, “Structure-measure: A new way to evaluate foreground maps,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 4548–4557, 2017.
- [12] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, “Enhanced-alignment measure for binary foreground map evaluation,” in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 698–704, 2018.
- [13] R. Margolin, L. Zelnik-Manor, and A. Tal, “How to evaluate foreground maps?,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 248–255, 2014.
- [14] K. Shen, X. Zhou, and Z. Liu, “MINet: Multiscale interactive network for real-time salient object detection of strip steel surface defects,” *IEEE Transactions on Industrial Informatics*, 2024.
- [15] X. Zhou, K. Shen, and Z. Liu, “ADMNet: Attention-guided densely multi-scale network for lightweight salient object detection,” *IEEE Transactions on Multimedia*, 2024.
- [16] M. S. Lee, W. Shin, and S. W. Han, “TRACER: Extreme attention guided salient object tracing network,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pp. 12993–12994, 2022.
- [17] G. Li, Z. Liu, X. Zhang, and W. Lin, “Lightweight salient object detection in optical remote-sensing images via semantic matching and edge alignment,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–11, 2023.
- [18] W. Cui, K. Song, H. Feng, X. Jia, S. Liu, and Y. Yan, “Autocorrelation-aware aggregation network for salient object detection of strip steel surface defects,” *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–12, 2023.
- [19] J. Zhu, X. Qin, and A. El-Saddik, “DC-Net: Divide-and-conquer for salient object detection,” *Pattern Recognition*, vol. 157, p. 110903, 2025.
- [20] X. Liu and L. Wang, “MSRMNet: Multi-scale skip residual and multi-mixed features network for salient object detection,” *Neural Networks*, vol. 173, p. 106144, 2024.
- [21] H. Luo, J. Wang, and B. Liang, “Spatial attention feedback iteration for lightweight salient object detection in optical remote sensing images,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024.
- [22] Y. Liu, L. Zhou, G. Wu, S. Xu, and J. Han, “TCGNet: Type-correlation guidance for salient object detection,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 6633–6644, 2023.
- [23] G. Li, Z. Bai, Z. Liu, X. Zhang, and H. Ling, “Salient object detection in optical remote sensing images driven by transformer,” *IEEE Transactions on Image Processing*, vol. 32, pp. 5257–5269, 2023.
- [24] Y. Yi, N. Zhang, W. Zhou, Y. Shi, G. Xie, and J. Wang, “GPONet: A two-stream gated progressive optimization network for salient object detection,” *Pattern Recognition*, vol. 150, p. 110330, 2024.